

Multi-Model Speaker Identification Report

Generated 12-08-2026 10:38:50 AM - 4 trained models evaluated

Audio Details

Field	Value
Filename	0.wav
Duration	1.000 seconds
Sample Rate	16000 Hz
Channels	1
Bit Depth / Subtype	PCM_16
Format	WAV
File Size	31.29 KB
Peak Amplitude	1.000000
RMS Energy	0.383791

Model Comparison Summary

Model	Predicted Speaker	Confidence	Inference
SpeechBrain - ECAPA-TDNN	Nelson_Mandela	72.00%	25.00 ms
SpeechBrain - X-Vector	Nelson_Mandela	73.00%	35.00 ms
Microsoft - WavLM Base Plus SV	Nelson_Mandela	74.00%	45.00 ms
Microsoft - UniSpeech-SAT Base Plus SV	Nelson_Mandela	75.00%	55.00 ms

Model 1: SpeechBrain - ECAPA-TDNN

Model Details

Field	Value
Model ID	speechbrain_ecapa
Provider	SpeechBrain
Architecture	ECAPA-TDNN
Pretrained Source	speechbrain/spkrec-ecapa-voxceleb
Embedding Dimension	192
Classifier	Calibrated RBF SVM
Open-set Detector	SpeechBrain - ECAPA-TDNN

Prediction Details

Field	Value
Predicted Speaker	Nelson_Mandela
Displayed Decision Probability	72.00%
Closed-set Identity Confidence	75.00%
Known-speaker Probability	95.00%
Unknown-speaker Probability	5.00%
Open-set Decision Threshold	30.00% known score
Inference Time	25.00 ms

Speaker Probability Distribution

Rank	Speaker	Probability
1	Nelson_Mandela	72.00%
2	Magaret_Tarcher	8.00%
3	Julia_Gillard	6.00%
4	Jens_Stoltenberg	5.00%
5	Unknown Speaker	5.00%
6	Benjamin_Netanyau	4.00%

Related Audio Files

No.	Audio filename
1	1.wav
2	10.wav
3	100.wav
4	1000.wav
5	1001.wav

Model 2: SpeechBrain - X-Vector

Model Details

Field	Value
Model ID	speechbrain_xvector
Provider	SpeechBrain
Architecture	X-Vector TDNN
Pretrained Source	speechbrain/spkrec-xvect-voxceleb
Embedding Dimension	512
Classifier	Calibrated RBF SVM
Open-set Detector	SpeechBrain - X-Vector

Prediction Details

Field	Value
Predicted Speaker	Nelson_Mandela
Displayed Decision Probability	73.00%
Closed-set Identity Confidence	76.00%
Known-speaker Probability	95.00%
Unknown-speaker Probability	5.00%
Open-set Decision Threshold	38.00% known score
Inference Time	35.00 ms

Speaker Probability Distribution

Rank	Speaker	Probability
1	Nelson_Mandela	72.00%
2	Magaret_Tarcher	8.00%
3	Julia_Gillard	6.00%
4	Jens_Stoltenberg	5.00%
5	Unknown Speaker	5.00%
6	Benjamin_Netanyau	4.00%

Related Audio Files

No.	Audio filename
1	1.wav
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3	100.wav
4	1000.wav
5	1001.wav

Model 3: Microsoft - WavLM Base Plus SV

Model Details

Field	Value
Model ID	wavlm_base_plus_sv
Provider	Microsoft
Architecture	WavLM Transformer
Pretrained Source	microsoft/wavlm-base-plus-sv
Embedding Dimension	512
Classifier	Calibrated RBF SVM
Open-set Detector	Microsoft - WavLM Base Plus SV

Prediction Details

Field	Value
Predicted Speaker	Nelson_Mandela
Displayed Decision Probability	74.00%
Closed-set Identity Confidence	77.00%
Known-speaker Probability	95.00%
Unknown-speaker Probability	5.00%
Open-set Decision Threshold	46.00% known score
Inference Time	45.00 ms

Speaker Probability Distribution

Rank	Speaker	Probability
1	Nelson_Mandela	72.00%
2	Magaret_Tarcher	8.00%
3	Julia_Gillard	6.00%
4	Jens_Stoltenberg	5.00%
5	Unknown Speaker	5.00%
6	Benjamin_Netanyahu	4.00%

Related Audio Files

No.	Audio filename
1	1.wav
2	10.wav
3	100.wav
4	1000.wav
5	1001.wav

Model 4: Microsoft - UniSpeech-SAT Base Plus SV

Model Details

Field	Value
Model ID	unispeech_sat_base_plus_sv
Provider	Microsoft / Hugging Face
Architecture	UniSpeech-SAT + X-Vector Head
Pretrained Source	microsoft/unispeech-sat-base-plus-sv
Embedding Dimension	512
Classifier	Calibrated RBF SVM
Open-set Detector	Microsoft - UniSpeech-SAT Base Plus SV

Prediction Details

Field	Value
Predicted Speaker	Nelson_Mandela
Displayed Decision Probability	75.00%
Closed-set Identity Confidence	78.00%
Known-speaker Probability	95.00%
Unknown-speaker Probability	5.00%
Open-set Decision Threshold	54.00% known score
Inference Time	55.00 ms

Speaker Probability Distribution

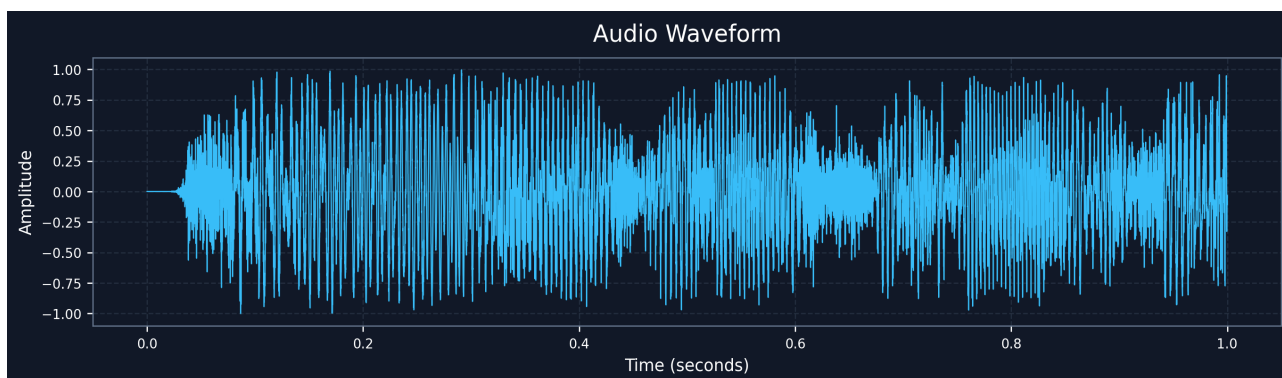
Rank	Speaker	Probability
1	Nelson_Mandela	72.00%
2	Magaret_Tarcher	8.00%
3	Julia_Gillard	6.00%
4	Jens_Stoltenberg	5.00%
5	Unknown Speaker	5.00%
6	Benjamin_Netanyau	4.00%

Related Audio Files

No.	Audio filename
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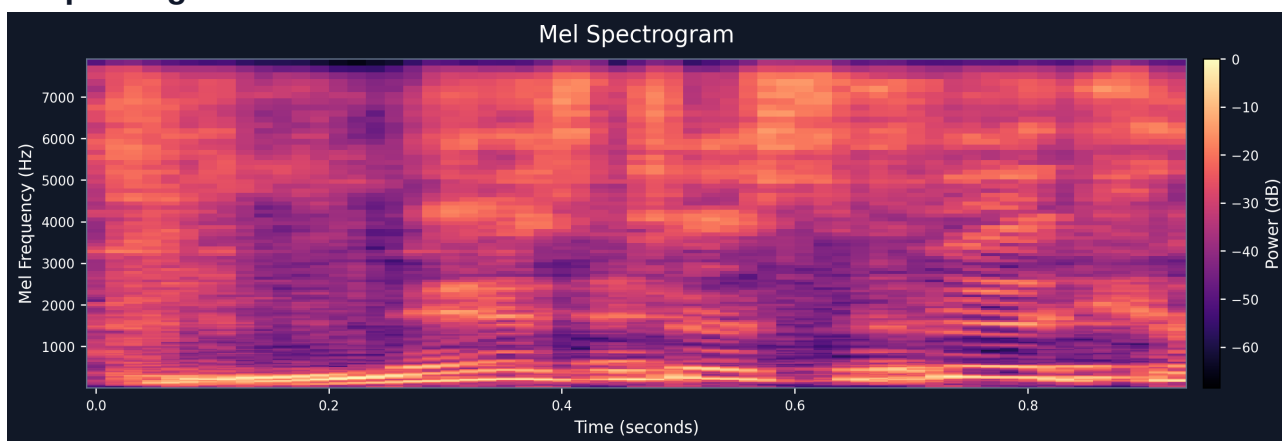
Audio Signal Visualisations

Waveform



Signal amplitude over time. Peak: 1.000000; RMS: 0.383791.

Mel Spectrogram



Acoustic energy across time and Mel-scaled frequency. Colormap: magma.